Mounika Geriki

Data Scientist / Machine Learning Engineer
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PROFESSIONAL SUMMARY

Senior AI/ML and Data Science professional with 4+ years of experience transforming large-scale data into actionable insights and production-ready ML systems. Skilled in **statistical analysis, predictive modeling, A/B testing, and machine learning**, with hands-on experience in **deep learning, NLP, and computer vision**. Delivered measurable impact including **94% model accuracy and 35% business growth metrics**, with strong expertise in **end-to-end ML lifecycle, data pipelines, and cloud-based deployment**.

TECHNICAL SKILLS

- **Programming Languages**: Python, SQL, Java, Scala, C++, R, JavaScript, Bash, Go, MATLAB
- **Machine Learning & AI**: TensorFlow, PyTorch, Scikit-learn, Keras, XGBoost, LightGBM, OpenCV, NLTK, spaCy, Hugging Face Transformers, MLflow, Kubeflow, AutoML, DSPy, LangGraph
- **Deep Learning**: CNN, RNN, LSTM, GRU, Transformer, BERT, GPT, Computer Vision, Natural Language Processing, Reinforcement Learning, Generative AI
- **LLM & Generative AI:** RAG Pipelines, Vector Databases (FAISS, Pinecone), Prompt Engineering, LLM Evaluation (RAGAS, TruLens), Agent Frameworks (LangChain, AutoGen), OpenAI & Gemini APIs
- **Big Data & Cloud**: Apache Spark, Kafka, Hadoop, AWS (SageMaker, EC2, S3, Lambda, CloudWatch), GCP (BigQuery, Vertex AI, Cloud Functions), Azure ML, Databricks
- **MLOps & DevOps**: Docker, Kubernetes, Jenkins, Git, CI/CD, Model Versioning, A/B Testing, Monitoring, Logging, Automated Testing, Infrastructure as Code, Azure DevOps, GitLab, Model Governance, Model Serving (FastAPI, TorchServe), Drift Monitoring, Feature Store (Feast)
- **Data Analysis**: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Tableau, Power BI, Excel, Statistical Analysis, Data Visualization, Feature Engineering
- **Data Engineering:** DBT Modeling, Delta Lake Optimization, Medallion Architecture (Bronze–Silver–Gold), Slowly Changing Dimensions (Type 1 & 2), Change Data Capture (CDC) Pipelines, Automated Data Quality Validation
- **Databases**: MySQL, PostgreSQL, MongoDB, Redis, ElasticSearch, Neo4j, Data Warehousing, ETL Pipelines, Data Mode ling

PROFESSIONAL WORK EXPERIENCE

Stripe | AI/ML Engineer| San Francisco, California, USA. **Oct 2024 – Present**

Project: GenAI-Powered Payment Intelligence Platform

Responsibilities:

- Designed and deployed **AI models** for **payment fraud detection**, **anomaly classification**, and **transaction scoring** using **PyTorch**, **TensorFlow**, and **XGBoost** on large-scale financial datasets.
- Fine-tuned **open-source LLMs (LLaMA 2, Falcon, GPT-NeoX)** with **LoRA** and **PEFT** to generate context-aware insights for **transaction disputes** and **merchant communications**.
- Built **retrieval-augmented generation (RAG)** workflows using **LangChain**, **FAISS**, and **Pinecone** to enable **natural-language querying** of historical payments and compliance data.
- Developed **multimodal ML pipelines** that combined structured transaction data with textual merchant metadata, improving **fraud-detection recall by over 20%**.
- Engineered **real-time data-processing and inference pipelines** with **Kafka**, **PySpark**, and **AWS Kinesis** to analyze **10 M+ transactions daily** with sub-second latency.
- Automated model training, evaluation, and deployment through **AWS SageMaker**, **MLflow**, **Docker**, and **GitHub Actions**, ensuring continuous integration and versioned retraining.
- Integrated **AI copilots** and **microservices** via **FastAPI** and **AWS Bedrock** to assist analysts with dispute resolution and risk investigations using natural-language prompts.
- Applied **explainable-AI frameworks** (**SHAP**, **LIME**, **Captum**) to interpret model outputs and satisfy regulatory transparency and audit requirements.
- Partnered with platform and security teams to establish **LLMOps standards** for bias monitoring, data lineage, and responsible- AI governance.
- Collaborated cross-functionally with Data Science, Risk, and Infrastructure teams to **productionize AI models** and optimize inference workloads on **GPU clusters**.
- Delivered internal workshops on **Generative-AI adoption**, **prompt-engineering**, and **scalable model-serving architectures.**

TCS | Data Scientist | India. **Sep 2021 – Jan 2024**
Project: Fraud Detection & Compliance Analytics
Responsibilities:

- Performed **exploratory data analysis (EDA)** on **millions of transaction records** using SQL and PySpark, identifying fraud patterns, anomalies, and key behavioral trends.
- Engineered **high-impact features** (transaction frequency, velocity, network relationships) to improve model performance and fraud detection accuracy.
- Built and evaluated **supervised and unsupervised models** (Logistic Regression, Random Forest, XGBoost, Autoencoders) to classify fraudulent transactions and detect anomalies.
- Conducted **model evaluation and validation** using cross-validation, ROC-AUC, precision-recall, and confusion matrix analysis to optimize performance on imbalanced datasets.
- Developed **NLP models (BERT, RoBERTa)** to classify AML/KYC documents and extract key entities, improving automation in compliance workflows.
- Applied **graph analytics and network-based modeling** to uncover hidden relationships between accounts and detect coordinated fraud patterns.
- Designed and analyzed **A/B testing and threshold optimization strategies**, reducing false positives by **28%** while improving anomaly detection recall by **19%**.
- Optimized large-scale data systems using **Delta Lake (25TB+)**, improving query performance by **38%** and accelerating analytics.
  Built **interactive dashboards (Tableau, Python)** to monitor fraud trends, model performance, and KPIs, enabling data-driven decision-making for stakeholders.
- Applied **Explainable AI techniques (SHAP, LIME)** to interpret model predictions and communicate insights to risk and compliance teams.
- Collaborated with cross-functional teams to translate business problems into **data-driven solutions**, improving fraud detection efficiency and reducing manual review effort by **40%.**

Wipro | Machine learning engineer intern | India. **Oct 2020 – Jun 2021**
Responsibilities:

- Performed **EDA and data preprocessing** on large structured and unstructured datasets using Python (Pandas, NumPy) and SQL, improving data quality and feature usability%.
- Developed **machine learning models (classification & regression)** using Scikit-learn and XGBoost to support business forecasting and customer analytics use cases.
- Built **NLP pipelines (NLTK, spaCy)** for text classification and sentiment analysis, improving model accuracy and enabling automated insights extraction.
- Assisted in developing **ETL pipelines** using Python and SQL to ingest, clean, and transform data from multiple sources into analytics-ready formats.
- Created **interactive dashboards (Tableau, Power BI)** to track KPIs and support data-driven decision-making.
- Supported **A/B testing and statistical analysis**, helping evaluate product features and business strategies.
- Gained hands-on exposure to **cloud platforms (AWS – S3, EC2)** and version control using Git.
- Collaborated with cross-functional teams in an **Agile/Scrum environment**, contributing to documentation and sprint deliverables.

EDUCATION

Master of Science - Data Science from Stony Brook University - Stony Brook, NY

Bachelor of Engineering Electronics and Communication Engineering from Bangalore Institute of Technology - India